

2.4 Graded Linear Series and Families of Ideals

Def 2.4.1 (Graded linear series) X irre variety

A graded linear series on X consists of a collection

$$V_{\bullet} = \{V_m\}_{m \geq 0}$$

of subspaces $V_m \subseteq H^0(X, \mathcal{O}_X(mD))$, D integral divisor

stf $V_k \cdot V_\ell \subseteq V_{k+\ell}$ for all $k, \ell \geq 0$

where $V_k \cdot V_\ell = \text{Im} (H^0(X, \mathcal{O}_X(kD)) \otimes H^0(X, \mathcal{O}_X(\ell D)) \rightarrow H^0(X, \mathcal{O}_X(k+\ell D)))$

It is also required that V_0 contain all constant functions.

Given a graded linear series $V_{\bullet} = \{V_m\}$, set

$$R(V_{\bullet}) = \bigoplus_{m=0}^{\infty} V_m$$

by definition, $R(V_{\bullet})$ is a graded $\mathbb{C}$ -subalgebra of $R(D) = R(X, D)$

We call $R(V_{\bullet})$ the section ring or graded ring associated to $V_{\bullet}$.

Def 2.4.2. $V.$ on a proj variety X is f.g. if $R(V.)$ is f.g. as $\mathbb{C}$ -alg.

Ex 2.4.3.

(ii) $\mathcal{I} \subseteq \mathcal{O}_X$ be the ideal sheaf. $c > 0$ $\mathbb{R}$, set

$$V_m = H^0(X, \mathcal{O}_X(mD) \otimes \mathcal{I}^{\lceil mc \rceil})$$

Since $\lceil ac \rceil + \lceil bc \rceil \geq \lceil (a+b)c \rceil$, $V.$ is a graded linear series.

which we denote by $T.(X, \mathcal{O}_X(D) \otimes \mathcal{I}^c)$

Sometimes, even when $R(D)$ is f.g., $T.(X, \mathcal{O}_X(D) \otimes \mathcal{I}^c)$ may fail to be

$X = \mathbb{P}^1$, $D = \mathcal{O}(1)$, $R(D) = \bigoplus_{m \geq 0} H^0(\mathbb{P}^1, \mathcal{O}(m)) \cong \mathbb{C}[x, y]$ f.g. as $\mathbb{C}$ -algebra.

Take $\mathcal{I} = m_x$, $c = \sqrt{2} - 1$, $V_m = H^0(X, \mathcal{O}_X(m) \otimes m_x^{\lceil mc \rceil})$

考虑 $S_m = x^{\lceil mc \rceil} y^{m - \lceil mc \rceil} \in V_m$, 我们说明 S_m 不能由低项生成对无限多 m .

即考虑 $a+b=m$ for every a, b , 希望 $\lceil ac \rceil + \lceil bc \rceil > \lceil mc \rceil$ (*)

如果这样的 m 有无限个, 则说明 $R(V.)$ 不是有限生成

$$(*) \Leftrightarrow \lfloor ac \rfloor + \lfloor bc \rfloor + 1 > \lfloor mc \rfloor = \lfloor ac \rfloor + \lfloor bc \rfloor + \{ac\} + \{bc\}$$

$$\Leftrightarrow \{ac\} + \{bc\} < 1$$

$$\text{Since } b = m - a \Leftrightarrow \{ac\} + \{mc - ac\} < 1$$

If $|mc| \geq |ac|$ ✓

If $|mc| < |ac|$ for some a , then $|ac| + |mc - ac| = 1 + |mc|$ 矛盾

从而 $(*) \Leftrightarrow |mc| \geq |ac|$ for all $a < m$

而易知存在无穷多个这样的 m .

iii) Fix $\chi \in \mathbb{P}^n$, def $V_m = \{s \in H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m)) \mid s_\chi = 0\}$,

由于 $s_\chi = 0$, 假设 $\chi = [1:0:\dots:0]$, 从而不能有 χ_0^m 项, 考虑 $s_m = \chi_0^{m-1} \chi_1$

任意 $f \in V_a, g \in V_b, a+b=m$ the highest deg of χ_0 in fg is $m-2$

$\Rightarrow V$ is not finitely generated.

iv) Suppose $X \subseteq Y$, $p = D'/X$, D' is a divisor on Y . X, Y varieties

Def $V_m = \text{Im}(H^0(Y, \mathcal{O}_Y(mD')) \rightarrow H^0(X, \mathcal{O}_X(mD)))$

$V = \{V_m\}$ forms a graded linear series, denoted by $\text{Tr}_X(D')$.

用来刻画可以提升的截面

接下来我们介绍对于分次线性系的三种操作

1. Product of two graded linear series

Given $V. \sim D$, $W. \sim E$, we can define

$$(V. \cdot W.)_m = V_m \cdot W_m \subseteq H^0(X, \mathcal{O}_X(mD + m\bar{E}))$$

by the image of $V_m \otimes W_m$ under the multiplication

$$H^0(X, \mathcal{O}_X(mD)) \otimes H^0(X, \mathcal{O}_X(m\bar{E})) \rightarrow H^0(X, \mathcal{O}_X(mD + m\bar{E}))$$

2. Intersection and span of graded linear series

Given $V., W. \sim D$, we define two new graded linear series

$$(V. \cap W.)_m = V_m \cap W_m$$

$$\text{Span}(V., W.)_m = \overline{\sum_{k+l=m} V_k \cdot W_l}$$

3. (Veronese of graded linear series)

If $V. \sim D$, $p \in \mathbb{Z}^+$, then define $V.^{(p)}$ by

$$(V.^{(p)})_m = V_{mp} \subseteq H^0(X, \mathcal{O}_X(mppD))$$

(半群, 保证平滑增长)

Thm: Let $R = \bigoplus_{m \geq 0} R_m$ be a graded ring and $d > 0$, if R is an ID,

R_0 is a Noetherian ring, then R is f.g. $\Leftrightarrow R^{(d)} = \bigoplus_{m \geq 0} R_{md}$ is f.g.

证明: " $\Leftarrow$ " If $R^{(d)}$ is f.g. as an R_0 -algebra.

Since R_0 noetherian $\Rightarrow R^{(d)}$ noetherian, then we have

$$R = \bigoplus_{i=0}^{d-1} R^{(d,i)}, \quad R^{(d,i)} = \bigoplus_{k \geq 0} R_{i+kd}, \quad R^{(d,0)} = R^{(d)}$$

for $R^{(d,i)} \neq 0$, take $z \in R^{(d,i)}$, $\deg(z) = md+i$

Consider $a = z^{d-1}$ $\deg(a) \equiv -i \pmod{d}$

Since R is an ID, $a \neq 0$, we can define

$$\phi_i: R^{(d,i)} \rightarrow R^{(d)} \text{ by}$$

$$v \mapsto v \cdot a.$$

which is injective, $\Rightarrow R^{(d,i)} \cong \alpha_i \subseteq R^{(d)}$ as an ideal.

which is f.g. by the Noetherian of $R^{(d)}$.

Now, we have proved that $R^{(d,i)}$ is a f.g. $R^{(d)}$ -module.

Since $R = \bigoplus_{i=0}^{d-1} R^{(d,i)}$, then R is a f.g. R_0 -algebra.

$\Rightarrow$ Lemma: R is a f.g. R_0 -algebra $\Leftrightarrow R^+$ is a f.g. R -module.

Proof of lemma: Trivial.

Now, assume R is f.g. as an R_0 -algebra, we can suppose.

R is generated by $f_1, \dots, f_k$, $\deg(f_i) = d_i$

Then for every monomial $M = f_1^{a_1} \dots f_k^{a_k} \in R^{(d)}$, consider

$$a_i = q_i d + r_i \quad 0 \leq r_i < d$$

$$M = (f_1^d)^{q_1} \dots (f_k^d)^{q_k} \cdot f_1^{r_1} \dots f_k^{r_k} \Rightarrow R \text{ is a f.g. } R^{(d)}\text{-module.}$$

Since R_+ is a f.g. R -module $\Rightarrow R_+$ is a f.g. $R^{(d)}$ -module

$R_+ = \bigoplus R_+^{(d, i)}$, then $R_+^{(d, i)}$ is a f.g. $R^{(d)}$ -module for any i .

$\Rightarrow R_+^{(d)}$ is a f.g. $R^{(d)}$ -module $\Rightarrow R^{(d)}$ is a f.g. R_0 -algebra

$\Rightarrow$ Suppose $R = R_0[X_1, \dots, X_n]$, $\deg(X_i) = d_i$

For $x_1^{a_1} \cdots x_n^{a_n} \in R^{(d)}$, we have $a_1 d_1 + \cdots + a_n d_n \equiv 0 \pmod d$.

Consider $M = \{(c_1, \dots, c_n) \in \mathbb{N}^n \mid \sum c_i d_i \equiv 0 \pmod d\}$ which is a semigroup.

By Gordan's lemma, M is f.g. $\Rightarrow R^{(d)}$ is f.g. as an R_0 -algebra.

Proof of Gordan's lemma:

$$M \Leftrightarrow \left\{ \sum c_i d_i - kd = 0 \right\}_k$$

扩充解域至 $\mathbb{R}^{n+1}$, 可看成是一个多面体锥:
$$\begin{cases} x_1 d_1 + \cdots + x_n d_n - yd = 0 \\ x_i \geq 0, y \geq 0 \end{cases}$$

从而 $\left\{ \sum c_i d_i - kd = 0 \right\}_k$ 该线性方程组会截出来一个多面体锥 C .

M 即为 $C \cap \mathbb{Z}^n$, 此时 C 只有有限条棱 (有限个面锥).

并且由于系数为整数, 故可以在射线上取个全称为整数的向量, 记为 $v_1, \dots, v_r$.

从而任意点 $x \in C$ 可写为 $x = \alpha_1 v_1 + \cdots + \alpha_r v_r$ ($\alpha_i \in \mathbb{R}, \alpha_i \geq 0$)

若 $x \in M$, $x = \sum (\lfloor \alpha_i \rfloor + \{\alpha_i\}) v_i$

若记 $y = \sum \lfloor \alpha_i \rfloor v_i$, $z = \sum \{\alpha_i\} v_i = x - y$

由于 x, y 皆为整点, 从而 z 也为整点, 但 z 位于 $P = \{\sum \beta_i v_i \mid 0 \leq \beta_i \leq 1\}$ 中.

从而有限整点, 因此记 $S = P \cap \mathbb{Z}^n$, M 由 $\{v_i\}$ 与 S 有限生成.

Def 2.4.7. Let $V_\bullet$ be a graded linear series belonging to a divisor D . The semigroup of $V_\bullet$ is the set

$$\mathbb{N}(V_\bullet) = \{m \geq 0 \mid V_m \neq 0\}$$

The exponent $e(V_\bullet)$ of $V_\bullet$ is the greatest common divisor of all the elements in $\mathbb{N}(V_\bullet)$

Def 2.4.8. (Iitaka dimension of $V_\bullet$) If $V_m = 0$ for all m , put

$K(V_\bullet) = -\infty$. Otherwise, given a positive integer $m \in \mathbb{N}(V_\bullet)$, write

$$\phi_m: X \dashrightarrow \mathbb{P}H^0(X, V_m)$$

denote by $\phi_m(X)$ the closure of its image.

Assuming X is normal, the Iitaka dimension $K(V_\bullet)$ of $V_\bullet$ is defined to be $K(V_\bullet) = \max_{m \in \mathbb{N}(V_\bullet)} \{\dim \phi_m(X)\}$

Def 2.4.9. (Stable base locus of a graded linear series)

The stable base locus $IB(V_\bullet)$ of $V_\bullet$ is the algebraic subset

$$IB(V_\bullet) = \bigcap_{m \geq 0} B_S(|V_m|)$$

Ex 2.4.11. $V. \sim D$ on X . $\mathfrak{b}_m = \mathfrak{b}(V_m)$ be the base ideal of V_m . Then $\mathfrak{b}_k \mathfrak{b}_e \subseteq \mathfrak{b}_{k+e}$.

Def 2.4.12. X proj variety of dim n . $V. \sim D$.

Then we def $\text{vol}(V.) = \limsup_{m \in \mathbb{N}(V.)} \frac{\dim V_m}{m^n/n!}$

Example 2.4.13. Let D be the hyperplane divisor on $X = \mathbb{P}^n$, let $\mathfrak{m}_x$ be the ideal of a point $x \in \mathbb{P}^n$, for $0 < c < 1$, def $V. = T.(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1) \otimes \mathfrak{m}_x^c)$

Then $\text{vol}(V.) = 1 - c$.

计算: Suppose $x = [1:0:0:\dots:0]$. $\dim V_m \overset{\text{solution of}}{\iff} x_1 + x_2 + \dots + x_{n+1} =$

$$x_1 \geq \lceil mc \rceil, x_i \in \mathbb{N} \Rightarrow \dim V_m = \binom{m - \lceil mc \rceil + n}{n}$$

$$\text{vol}(V.) = \limsup_m \frac{(m - \lceil mc \rceil + n) \cdots (m - \lceil mc \rceil + 1)}{m^n} \leq 1 - c$$

Def 2.4.14 (Graded family of ideals). X irre variety.

A graded family or graded system of ideals $\alpha. = \{\alpha_m\}$ on X is a collection of ideal sheaves $\alpha_i \subseteq \mathcal{O}_X$ s.t.

$$\alpha_k \cdot \alpha_\ell \subseteq \alpha_{k+\ell}.$$

for all $k, \ell \geq 1$, $\alpha_0 = \mathcal{O}_X$.

Def 2.4.15. The Rees algebra of $\alpha.$ is the graded $\mathcal{O}_X$ -algebra

$$\text{Rees}(\alpha.) = \bigoplus_{m \geq 0} \alpha_m$$

We say that $\alpha.$ is f.g. if $\text{Rees}(\alpha.)$ is f.g. as an $\mathcal{O}_X$ -algebra

i.e. $\exists m_0$ s.t. $\text{Rees}(\alpha.)$ is generated as an $\mathcal{O}_X$ -algebra by its terms

of degree $\leq m_0$. (这意指 $\forall x \in X, \exists U, f_1, \dots, f_n \in \text{Rees}(\alpha.)(U)$ s.t. f_i generate stalks for every $y \in U$. 并且这要求 X 连通)

下面我们看一些例子

Example 2.4.16. X irre variety

(i) Given an ideal sheaf $\mathfrak{b} \subseteq \mathcal{O}_X$, $\text{Pow.}(\mathfrak{b}) = \{\alpha_m = \mathfrak{b}^m\}$

We will see that any f.g. graded system is essentially of this form.

(ii) Let $V_n \sim D$ on X . Then the base ideals $\mathfrak{b}_m = \mathfrak{b}(|V_m|)$ form a graded family $\mathfrak{b}(V_n)$.

证明: Since $V_n \cdot V_m \subseteq V_{n+m}$

$$\Rightarrow \mathfrak{b}(V_n) \cdot \mathfrak{b}(V_m) \subseteq \mathfrak{b}(V_{n+m})$$

locally, $\mathcal{O}_X(D) = L \xleftarrow{\cdot e} \mathcal{O}_X$ $V_n \nsubseteq \mathfrak{b}_n \cdot e^n, V_m \nsubseteq \mathfrak{b}_m \cdot e^m, f \in \mathfrak{b}(V_n), g \in \mathfrak{b}(V_m)$

(V) Assume X is normal, $\mu: X' \rightarrow X$ be a projective birational morphism

Let D be an effective μ -exc Cartier divisor, then

$$\alpha_m = \mu_* \mathcal{O}_{X'}(-mD) \subseteq \mathcal{O}_X \quad (\text{by Zariski's main thm } \mu_* \mathcal{O}_{X'} = \mathcal{O}_X)$$

form a graded family.

$$\alpha_k \cdot \alpha_\ell \subseteq \alpha_{k+\ell} \quad \text{deduced by}$$

$$\mu_* \mathcal{O}_{X'}(-kD) \otimes \mu_* \mathcal{O}_{X'}(-\ell D) \rightarrow \mu_* \mathcal{O}_{X'}(-(k+\ell)D)$$

Nadel Vanishing: X sm proj variety over $\mathbb{C}$, $D \geq 0$ $\mathbb{Q}$ -divisor

L is a line bundle on X . If $L-D$ is nef and big, we have

$$H^i(X, \mathcal{O}_X(kL + D) \otimes \mathcal{I}(D)) = 0 \quad \text{for all } i > 0$$

$$J(D) = \pi_{X*} \mathcal{O}_Y (K_{Y/X} - L^* \tilde{D}_1)$$

Example 2.4.18 (Graded systems from valuations).

Suppose v is a valuation on the function field $\mathbb{C}(x)$ taking values in $\mathbb{R}$ and $v(f) \geq 0$ for regular function f .

Then we define the valuation ideals

$$\alpha_m = \{f \in \mathcal{O}_X \mid v(f) \geq m\}$$

form a graded system of ideal sheaves,

For non-divisorial valuations, there are some interesting examples.

Assume $X = \mathbb{C}^2$, $\mathbb{C}(X) = \mathbb{C}(x, y)$.

(1) Let $p(t) = \sum_{i=1}^{\infty} \frac{t^i}{i!} \in \mathbb{C}[[t]]$, define

$$v(f) = \text{ord}_t f(t, p(t))$$

for $f \in \mathbb{C}[x, y]$. The corresponding valuation ideals are given by

$$\alpha_m = (x^m, y - p_{m-1}(x))$$

显然不是有限生成的. $y - p_{m-1}(x)$ 会一直是新的生成元.

(ii) Valuation by irrational weighted degree.

Let $\alpha_m \subseteq \mathbb{C}[x, y]$ be the monomial ideal spanned by all monomials $x^i y^j$ with

$$i + j\sqrt{2} \geq m$$

其赋值函数为 v . $v(x) = 1$, $v(y) = \sqrt{2}$. 同样不是有限生成的, 类比 2.4.3.

Def 2.4.24

Given $\alpha. = \{\alpha_m\}$, $\beta. = \{\beta_m\}$ on X , define

$$(\alpha. \wedge \beta.)_m = \alpha_m \wedge \beta_m \quad \text{Intersection}$$

$$(\alpha. \cdot \beta.)_m = \alpha_m \cdot \beta_m \quad \text{product}$$

$$(\alpha. + \beta.)_m = \sum_{k+l=m} \alpha_k \cdot \beta_l \quad \text{sum}$$

Similarly, we can define

Def 2.4.25 Let $\alpha. = \{\alpha_m\}$ be a graded system of ideals on an irreducible variety X . The semigroup of $\alpha.$ is

$$N(\alpha.) = \{m \geq 0 \mid \alpha_m \neq (0)\}$$

The exponent $e(\alpha.)$ is the greatest common divisor in $N(\alpha.)$

Def 2.4.26 (Veronese of a graded family)

Given $\alpha. = \{\alpha_m\}$ on X , the p^{th} Veronese of $\alpha.$ is the graded system $\alpha.^{(p)}$ defined by $(\alpha.^{(p)})_m = \alpha_{pm}$.

Prop 2.4.27. Let $\alpha. = \{\alpha_m\}$ be a f.g. graded family of ideals on X . Then there is an integer $p > 0$ such that

$$\alpha.^{(p)} = \text{Pow.}(\alpha_p)$$

证明: Suppose $X = U_1 \cup U_2 \cup \dots \cup U_k$

If we can prove it for the affine case, then

$$(\alpha_{mp_i})|_{U_i} = (\alpha_{p_i}|_{U_i})^m$$

Then we can take $p = \prod p_i$, $\alpha.^{(p)} = \text{Pow.}(\alpha_p)$

Now, assume X is affine. 从而问题变为代数的有限生成

Lemma: Let $R. = R_0 \oplus R_1 \oplus \dots$ is a graded ring, If $R.$ is f.g. as an R_0 -algebra, then there exists an integer p s.t.

$R.^{(p)}$ is generated by R_p as an R_0 -algebra.

Proof of lemma: Suppose $x_1, \dots, x_n$ generate $R.$, $\deg(x_i) = d_i$

let $L = \text{lcm}(d_1, \dots, d_n)$. Consider $M = x_1^{a_1} \dots x_n^{a_n}$

希望证明 存在一个 $p > 0$, 使得对任意次数为 mp 的单项式 M , 都能提一个因子为 p 的因子

取 $p = nL$, n 为生成元个数.

Suppose $\deg M = mp$. i.e. $\sum a_i d_i = mp = mnL$, $m \geq 2$.

$$a_i d_i = q_i L + r_i \quad 0 \leq r_i < L$$

Since $d_i | L$, we have $d_i | r_i$.

Rewrite $\sum_{i=1}^n (q_i L + r_i) = mnL = L(\sum q_i) + \sum r_i$

Since $r_i < L \Rightarrow \sum r_i < nL = p$, so

$$(\sum q_i)L = mnL - \sum r_i > mnL - nL = (m-1)nL$$

$$\Rightarrow \sum q_i > n \quad (\text{by } m \geq 2)$$

So, we can take $0 \leq c_i \leq q_i$, s.t.

$$\sum c_i = n \quad y_i = x_i^{\frac{L}{d_i}}$$

Then $M = x_1^{a_1} \cdots x_n^{a_n} = y_1^{c_1} \cdots y_n^{c_n} M'$, $\deg \frac{M}{M'} = (\sum c_i) \cdot L = nL = p$

$\Rightarrow R^{(p)}$ is generated over R_0 by R_p .